THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5 & 6: **Prime Time, Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 2	two factors
2	**(b)** 14 and 15	HCF is 1
3	**(b)** the sum of its digits is divisible by 3	digit sum rule
4	**(d)** 4	rule
5	**(c)** 2	e.g. 11 and 13
6	**(b)** $2(l+b)$	sum of all sides
7	**(c)** $\frac{1}{2}\times$ base × height	half of a rectangle
8	**(c)** 26 cm	$2(8+5)$
9	**(b)** 21 cm	3×7
10	**(d)** 8 m	$72 \div 9$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The difference between the sum of alternate digits is 0 or a multiple of 11.	1+1
12	Digit sum 10 is not divisible by 3, so no.	1+1
13	$3 \times 10{,}000 = 30{,}000$ cm².	1+1
14	Fencing needs the perimeter; tiling needs the area.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	9, 18, 27, 36, 45, 54, 63, 72 (2); digit sums are 9 (1).	2+1
16	25,000 cm² (1.5); 5 m² (1.5).	1½+1½
17	143 = 11 × 13 (3).	1+2
	OR: LCM 12 + 1 = 13 (3).	1+2
18	Side 12 cm, area 144 cm² (1.5); breadth 10 cm, area 140 cm² (1.5).	1½+1½
	OR: $\frac{1}{2} \times 12 \times h = 54$, so $h = 9$ cm (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 300 (2). (ii) 300 (1). (iii) Divisible by all (2).	2+1+2
	OR: (i) 12 ways (6 pairs) (1). (ii) 1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84 (1). (iii) 83 has only two factors (1). (iv) Prime (2).	1+1+1+2
20	(i) 180 m² (1). (ii) 66 m × 30 = ₹1,980 (2). (iii) 180 plants (1). (iv) Surface covered (1).	1+2+1+1
	OR: (i) Shape changes perimeter (1). (ii) 50, 28, 22, 20 cm (3). (iii) 4 × 6 (1).	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 5 & 6: **Prime Time, Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** least common multiple	definition
2	**(a)** 97	99, 98 and 91 are not prime
3	**(d)** 4	common factors of 8 and 20
4	**(d)** 6	common factors of 18 and 30
5	**(c)** 24	smallest common multiple of 6 and 8
6	**(b)** 160 m	$2(50 + 30)$
7	**(d)** half the rectangle's area each	diagonal splits it
8	**(c)** 24 cm²	6×4
9	**(d)** 198 cm²	18×11
10	**(d)** 56 cm²	$\frac{1}{2} \times 16 \times 7$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Two primes with a difference of 2; 5 and 7.	1+1
12	It has only one factor.	1+1
13	$\frac{1}{2} \times 9 \times 4 = 18$ cm².	1+1
14	No; 6×2 and 4×4 have perimeter 16 but areas 12 and 16.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	108 (digit sum 9) (3).	1+2
16	Area counts squares covering the surface (1.5); perimeter measures boundary length (1.5).	1½+1½
17	It has 2 as a factor besides 1 and itself (3).	1+2
	OR: 100 → 10 × 10 → 2×5×2×5 = 2²×5² (3).	1+2
18	Perimeter 320 m (1.5); area 6,000 m² (1.5).	1½+1½
	OR: Length 14 cm; perimeter 42 cm (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) All other evens have 2 as a factor (2). (ii) No (1). (iii) 2 + 3 = 5 (2).	2+1+2
	OR: (i) LCM = 120 (2.5). (ii) 125 (2.5).	2½+2½
20	(i) 78 cm² (2). (ii) 34 − 6 + (6 + 3 + 3) = 40 cm (3).	2+3
	OR: (i) Approximate counting (1). (ii) Count full and half squares (2). (iii) Partial squares counted as halves (2).	1+2+2

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ANSWER KEY & MARKING SCHEME SET C

Chapters 5 & 6: **Prime Time, Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 29	no other factors
2	**(b)** prime factorisation	definition
3	**(a)** the sum of its digits is divisible by 9	digit sum rule
4	**(a)** both 2 and 3	6 = 2 × 3
5	**(d)** 17 and 19	differ by 2
6	**(a)** length × breadth	formula
7	**(c)** square centimetre (cm²)	square unit
8	**(a)** 36 m	4×9
9	**(a)** 10,000 cm²	100 cm × 100 cm
10	**(a)** may have different areas	perimeter ≠ area

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	23, 29, 31, 37.	1+1
12	Yes; last two digits 28 are divisible by 4.	1+1
13	39 cm.	1+1
14	$2(18 + 10) = 56$ cm.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	12 (3).	1+2
16	60 cm² (3).	1+2
17	223 is prime; 221 = 13 × 17; 225 = 15 × 15 (3).	1+2
	OR: 3 + 17 and 7 + 13 (3).	1+2
18	Side 9 cm; area 81 cm² (3).	1+2
	OR: Each 20 cm²; total 40 cm² (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) More than two factors (1). (ii) 20, 21, 22, 24, 25, 26, 27, 28 (2). (iii) Correct factorisations (2).	1+2+2
	OR: (i) Divisible by 11 (alternate digit sums 1+3 and 3+1 equal) (2). (ii) Not divisible (2). (iii) Alternate digit sum difference (1).	2+2+1
20	(i) 9,000 m² (1). (ii) 360 (1). (iii) 390 m × 20 = ₹7,800 (2). (iv) 0.9 ha (1).	1+1+2+1
	OR: (i) Diagram (1). (ii) 40 cm² each (2). (iii) Diagonal cuts it into two equal triangles (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

Chapters 5 & 6: **Prime Time, Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 2 × 2 × 3	all primes
2	**(c)** 4	2 × 2
3	**(b)** 5	common factors of 15 and 25
4	**(b)** 3	common factors of 9 and 12
5	**(d)** 15	smallest common multiple of 3 and 5
6	**(d)** area	surface
7	**(a)** rectangle	cut and move
8	**(c)** 84 cm²	12×7
9	**(b)** 100 cm²	25×4
10	**(a)** 110 cm²	$\frac{1}{2} \times 20 \times 11$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	1 and 17.	1+1
12	24.	1+1
13	Into two congruent triangles, each half the area.	1+1
14	24.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Not always; 12 is divisible by 4 and 6 but not 24 (3).	1+2
16	240 ÷ 4 = 60 tiles (3).	1+2
17	Digit sum 6 is divisible by 3; 123 = 3 × 41 (3).	1+2
	OR: Prime 7; composite 12; co-prime 8, 9; twin primes 11, 13 (3).	1+2
18	(1,36)=74; (4,9)=26; (6,6)=24 (3).	1+2
	OR: 10 cm (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Helps HCF, LCM (1). (ii) 252 = 2²×3²×7; 420 = 2²×3×5×7 (2). (iii) 84 (2).	1+2+2
	OR: (i) No divisor other than 1 and 13 (1). (ii) 91 = 7×13; 87 = 3×29 (2). (iii) Even numbers and multiples of 3, 5, 7 (2).	1+2+2
20	(i) Reasoning (2). (ii) 30 vs 60 (3).	2+3
	OR: (i) cm², m² and cm, m (1). (ii) 45,000 cm²; 80 m (2). (iii) Fencing, tiling etc. (2).	1+2+2

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